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Regression Prediction using AWS Machine Learning

- Posted by [Vozag](#) on June 7, 2015 at 9:30pm
- [View Blog](#)

We wanted to be able to predict median rent of a place given the median price of the home, median household income of the place and the percentage of homes vacant in that place. The data can be downloaded from [here](#)

The steps to be followed are

1. Create data source
2. Train the model
3. Evaluate the model
4. Generate predictions

To get started login into Amazon AWS console and click on machine learning. It shows all your entities by default. An entity can be an ML model, Data set, Evaluation etc

AWS Services Edit

Amazon Machine Learning

Entities

Create new... Actions

Refresh

Filter: All types

Entity name or ID

Items per page: 10

<< < 1 - 1 of 1 Entities > >>

	Name	Type	ID	Status	Creation Time
☐	spam-ham-training	Datasource	ds-cONjXWUBn3z	In progress	May 18, 2015 10:22:48 AM

<< < 1 - 1 of 1 Entities > >>

More info

[Getting Started Guide](#)
[Amazon Machine Learning Developer Guide](#)
[Machine Learning Concepts](#)
[API Reference](#)

Machine Learning Concepts

Amazon Machine Learning (Amazon ML) can solve business problems by finding and learning the patterns in your historical data and using the patterns to generate predictions. To get started, you provide Amazon ML with your data. Next, you use Amazon ML to train your ML model and evaluate the model's performance.

Datasource and ML model

[Getting Started with Amazon Machine Learning](#)

Creating a Datasource

To create a dataset click on “Create new” drop down button and then select “Data Source”

 Amazon Machine Learning ▾

Entities

Create new... ▾

Actions ▾

Datasource and ML model

Datasource

ML model

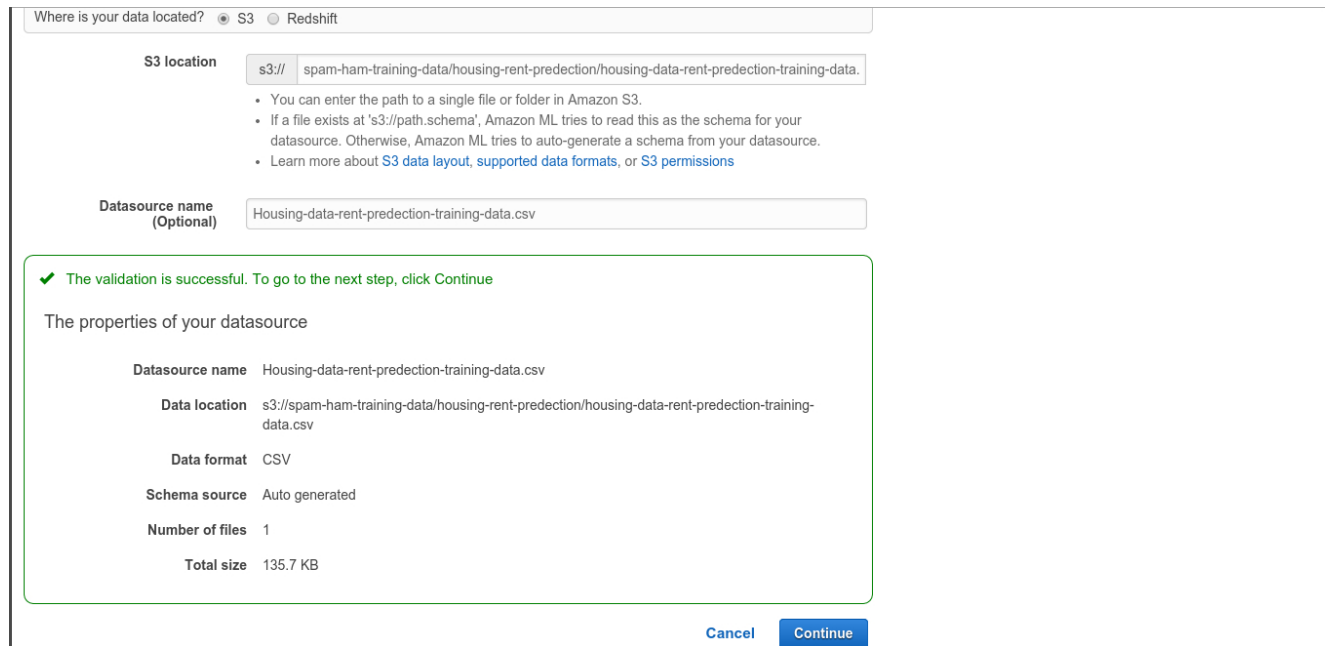
Evaluation

Batch prediction

Entity name or ID		Type
☐	Housing-data-with-house-hold-income-test...	Datasource
☐	Evaluation: ML model: Housing-data-with-...	Evaluation
☐	ML model: Housing-data-with-house-hold-i...	ML model

To create a Datasource, your data file needs to be present in either amazon S3 or RedShift

If you are getting data from S3, you need to provide the location of the data in your S3. Once you provide the information click “verify”



Where is your data located? ☒ S3 ☐ Redshift

S3 location

s3:// spam-ham-training-data/housing-rent-prediction/housing-data-rent-prediction-training-data.csv

- You can enter the path to a single file or folder in Amazon S3.
- If a file exists at 's3://path.schema', Amazon ML tries to read this as the schema for your datasource. Otherwise, Amazon ML tries to auto-generate a schema from your datasource.
- Learn more about [S3 data layout](#), [supported data formats](#), or [S3 permissions](#)

Datasource name (Optional)

Housing-data-rent-prediction-training-data.csv

✓ The validation is successful. To go to the next step, click Continue

The properties of your datasource

Datasource name	Housing-data-rent-prediction-training-data.csv
Data location	s3://spam-ham-training-data/housing-rent-prediction/housing-data-rent-prediction-training-data.csv
Data format	CSV
Schema source	Auto generated
Number of files	1
Total size	135.7 KB

[Cancel](#) [Continue](#)

The datasource is validated in this process

A *schema* is composed of all attributes in the input data and their corresponding data types. Amazon ML uses the information in the schema to correctly read and interpret the input data, compute statistics, apply the correct attribute transformations, and fine-tune its learning algorithms.

You can provide a separate schema file when you upload your AWS S3 data. Here we let Amazon ML to infer the attribute types and create a schema.

On the schema page check the “Does the first line in your CSV contain the column names?” option to “Yes”.

Make sure that the attributes in the file are assigned the correct datatype.

Schema

Amazon ML scanned your input data and inferred the column names and data type for each of the columns in your dataset. Review and edit as needed.

Does the first line in your CSV contain the column names? ☒ Yes ☐ No

ACTION: Change type (4) ▾

Search by attribute name 🔍

« < 1 - 7 of 7 > »					
☐	Name ▲	Data type ▴	Sample field value 1	Sample field value 2	Sample field value 3
☐	Geo_ID	Categorical ▾	05000US06001	05000US06003	05000US06005
☒	MedianHomeVa...	Numeric ▾	514900	371300	286500
☒	MedianHouseH...	Numeric ▾	71516	59931	53462
☒	MedianRent	Numeric ▾	1265	755	1039
☒	PercentageOfV...	Numeric ▾	0.2839175814	3.1404958678	2.6697006381
☐	PlaceName	Text ▾	Alameda County	Alpine County	Amador County
☐	state	Text ▾	ca	ca	ca

« < 1 - 7 of 7 > »

Cancel

Previous

Continue

Review the types properly and click continue

In the next page for “Do you want to use this dataset to create and/or evaluate a ML model?” choose “Yes”

This will let us select the target attribute

1. Input Data 2. Schema 3. Target 4. Row ID 5. Review

Target

Machine learning works by finding patterns that connect your data to the value to be predicted. To create an ML model, Amazon ML analyzes examples of data records with correct values. The column that contains these values in the training dataset is called the target.

Do you plan to use this dataset to create or evaluate an ML model? ☒ Yes ☐ No

Select the row containing the value you want to predict.

You have selected **MedianRent** as a target. ML models trained from this datasource will use **Numerical regression**.

Search by attribute name

Target	Name	Data type	Sample field value 1	Sample field value 2	Sample field value 3
☐	Geo_ID	Categorical	05000US06001	05000US06002	05000US06005
☐	MedianHouseV...	Numerical	514000	371300	286500
☐	MedianHouseH...	Numerical	71515	68831	59462
☒	MedianRent	Numerical	1285	755	1035
☐	PercentageOV...	Numerical	0.2839170014	3.1404868675	2.6607008381

Cancel Previous Continue

The “Target” is the attribute which the model must learn to predict. Here we want to predict Median rent of a place. So we select it as target

In the next page for “Do you want to select an identifier?” choose “Yes”

and in the next page check “Geo_ID”

and click on “Review”

1. Input Data 2. Schema 3. Target 4. Row ID 5. Review

Row identifier (optional)

A row identifier is used to let Amazon ML know which other column to include in the prediction output file. This column makes it easier to follow which prediction corresponds with which observation. This ID is for your reference only and will not affect the ML model training.

Do you want to select an identifier? ☒ Yes ☐ No

Geo_ID is selected as a record identifier. Click Finish to complete the wizard.

Search by attribute name

Row ID	Name	Data type	Sample field value 1	Sample field value 2	Sample field value 3
☒	Geo_ID	Categorical	05000US06001	05000US06003	05000US06005
☐	MedianHomeVa...	Numeric	514900	371300	286500
☐	MedianHouseH...	Numeric	71516	59931	53462
☐	PercentageOfV...	Numeric	0.2839175814	3.1404958678	2.6697008381
☐	PlaceName	Text	Alameda County	Alpine County	Amador County
☐	state	Text	ca	ca	ca

Cancel Previous Review

In the next page review the attributes and click “Finish”

Once you click finish you see the data source being “initialized”. It takes some time to reach “Completed” status

Training ML model

Amazon ML supports 3 types of ML models, namely

1. Binary classification
2. Multi class classification
3. Regression

The type of model depends on the type of data you want to predict

For binary classification AWS ML uses logistic regression algorithm and for multi class classification and regression, it uses multinomial logistic regression and linear regression algorithms respectively

Regression model

Since we want to predict the rent at a particular place, which is a number we use Regression ML. The ML model based on training data, computes one weight for each feature to form a model that can predict or estimate the target value

The learning algorithm consists of a [loss function](#) and an optimization technique. The loss function used for regression by AWS ML is [squared loss function](#) and the optimization technique is [SGD](#).

Create an ML model

You can create an ML model either from the datasource or from the “Create New” dropdown button in the dashboard, like you created the dataset

If you created it from the create new dropdown button you have to provide the name of the data source on which the model has to train

The screenshot shows the 'Create ML model' wizard in the Amazon Machine Learning console. The first step, 'Input data', is active. It provides instructions on locating data for training. Two options are available: 'I already created a datasource pointing to my S3 data' (selected) and 'My data is in S3, and I need to create a datasource'. A search bar for 'Datasource name or ID' is present. Below it, a table lists an existing datasource:

Datasource ID	Datasource name	Created on	Status
[REDACTED]	Housing-data-rent-prediction-training-data.csv	May 19, 2015 8:18:08 PM	Completed

'Cancel' and 'Continue' buttons are at the bottom right. The footer includes 'Feedback', 'English', and copyright information.

Click “Continue”. In the next page give the name of the model

In the next page for “Training and evaluation settings” choose “Default”
Because it is best to start with the simple and default options first.

By selecting this option an evaluation will automatically be generated. 70% of the data will be used for training and the remaining 30% will be used for evaluation

ML model settings

You can use the automatically suggested ML model settings, or you can choose to customize.

ML model Type REGRESSION

ML model target MedianRent

ML model name (Optional)

Training and evaluation settings

You can select additional settings for your ML model, or use the defaults provided by Amazon ML. Additionally, Amazon ML can reserve a portion of your input data and create an evaluation for your ML model.

☒ Default (Recommended)

Select this option to use the recommended ML model parameters and to have an evaluation automatically created for you. Amazon ML will split your input data and use 70 percent for training (training datasource) and 30 percent for evaluating this ML model (evaluation datasource).

Name this evaluation (Optional)

☐ Custom

Select this option to specify the available ML model settings and whether you would like to use a portion of your input data to evaluate this ML model.

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Evaluating the model

Once the model is built, it can be run on some data which it has not seen and the predicted values can be compared to that of the original value to evaluate the performance of the model.

Since we selected the “Default” option, an evaluation is automatically generated

For regression tasks, Root mean square error is used to evaluate the accuracy. The RMSE for our model is 278

ML model report

[Summary](#)[Settings](#)[Evaluations](#)

▼ Evaluation: ML mode...

[Summary](#)[Alerts \(0\)](#)[Explore performance](#)

Evaluation Summary

[Delete this Evaluation](#)

ID	[REDACTED]
Name	Evaluation: ML model: Housing-data-rent-predction-training-data.csv 
Datasource ID	[REDACTED]
Output location	Not available
Created on	05/19/15 20:28:10
Status	Completed
Message	Not available
Log	Download log

ML model performance

On your most recent evaluation, [REDACTED], the ML model's quality score is **better** than the baseline. 

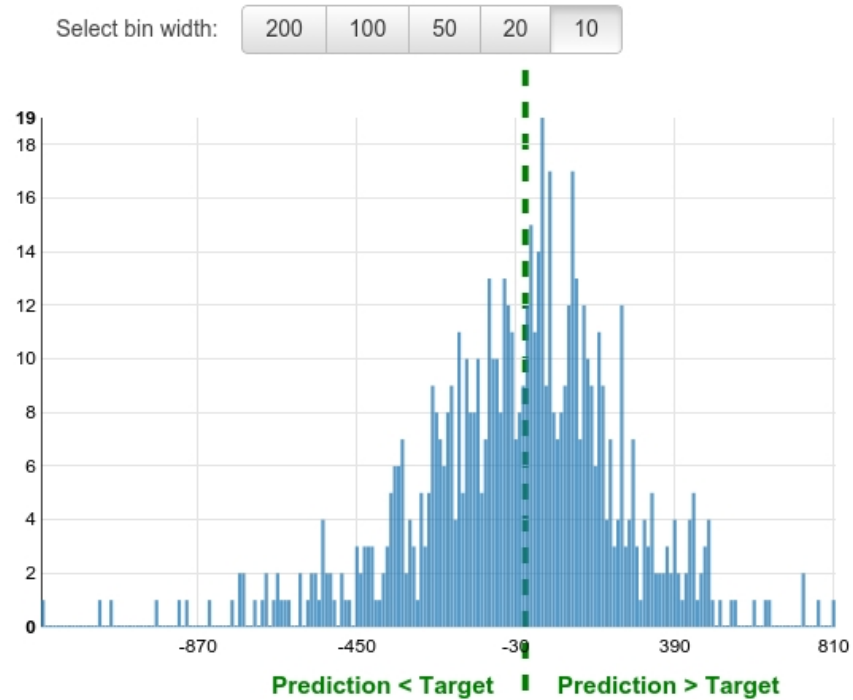
RMSE: 278.1663
RMSE baseline: 442.39
Difference: 164.22

[Explore model performance](#)

We can also see the the distribution of errors of the estimates. It can be seen by selecting Evaluations -> Explore Performance

ML model performance

This chart shows the distribution of the residuals of your ML model.



Generating Batch Predictions

You can start generating the batch predictions and real time predictions immediately

target medianrent

Evaluation

Number of evaluations 1

Latest evaluation result

[Perform another Evaluation](#)

Usage

A single dataset

Generate one-time predictions for a single dataset.

[Generate batch predictions](#)

Continuously in real time

Enable real time predictions.

[Enable for real time predictions](#)

sh

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You will need to create a datasource to generate the batch predictions

Cost

AWS Machine Learning charges an hourly rate for the compute time used to build predictive models, and then you pay for the number of predictions generated for your application. For real-time predictions you also pay an hourly reserved capacity charge based on the amount of memory required for your model.

For data analysis and model building amazon charges \$0.42 per Hour

For generating batch predictions \$0.10 per 1,000 predictions

For real time predictions \$0.0001 per prediction, rounded up to the nearest penny.

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Comment by [Dr. Dimitrios Geromichalos](#) on November 19, 2015 at 12:34am

Very interesting article. I'd like to add that from a risk management perspective the median is not the only relevant quantile. Here, also best and - especially worst - cases like the 5% and 95% are important. I did not see anything like that in ML yet, but I suppose the information could easily be obtained from the residuals. This quantile information is (still) the most important one for banks and insurers when they calculate figures like Value at Risk and Economic Capital and should be also useful for everyone who wants to minimize risks.

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A banner for Penn State Online's Graduate Certificate in Business Analytics. On the left, a man in a white shirt and tie is working at a desk with a laptop. The right side of the banner has a dark blue background with white text. The text reads "PENN STATE | ONLINE" in a large, serif font, followed by "Graduate Certificate in Business Analytics" in a smaller, sans-serif font. A blue button with white text and a play icon says "▶ GET STARTED".

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